In addition, for all k, k' and y, y', possibly equal, there exists an augmented connecting cycle $(k'_1, y'_1, \ldots, k'_R, y'_R)$, $(\widetilde{k}'_1, \widetilde{y}'_1, \ldots, \widetilde{k}'_R, \widetilde{y}'_R)$, such that $k'_1 = k$, $y'_1 = y$, and $\widetilde{k}'_r = k'$, $\widetilde{y}'_r = y'$ for some r.

(ii) *There exist finite sets of M values for y_1 and y_4 such that, for all r in $\{1, \ldots, R\}$, the matrices $A(y_r, \widetilde{y}_r, k_r, \widetilde{k}_r)$ and $A(y_{r+1}, \widetilde{y}_r, k_{r+1}, \widetilde{k}_r)$ have rank L, where $A(y, y', k, k')$ has (y_1, y_4) element*:

$$\Pr\big[Y_{i1} \leq y_1, Y_{i4} \leq y_4 | Y_{i2} = y, Y_{i3} = y', k_{i2} = k, k_{i3} = k', m_{i2} = 1\big].$$

We then have the following nonparametric identification result for the dynamic model under discrete worker heterogeneity.

THEOREM 2: *Let $T = 4$, and let Assumptions 2 and 4 hold. Suppose that firm classes are observed. Then, up to labeling of the types α:*

(i) *$G^f_{y_2,k\alpha}$ and $G^b_{y_3,k'\alpha}$ are identified for all (α, k, k'). Moreover, for all (k, y_2, k', y_3) for which there are job moves from (k, y_2) to (k', y_3), $p_{y_2 y_3, kk'}(\alpha)$ is identified for all α.*

(ii) *$F_{k\alpha}$ and $q_k(\alpha)$, and log-earnings cdfs in periods 3 and 4, are also identified. Last, type-specific transition probabilities between firm classes are identified.*

4. TWO-STEP GROUPED FIXED-EFFECTS ESTIMATION

In the previous section, we have provided conditions under which earnings distributions are identified in the presence of sorting and complementarities. These results hold at the firm class level k_{it}, where, in principle, k_{it} could coincide with the firm j_{it}. However, in matched employer employee panel data sets of typical sizes, estimating models with complementarities, dynamics, and two-sided heterogeneity may be ill-behaved due to the incidental parameter biases caused by the large number of firm-specific parameters that are solely identified from job movements. For this reason, we use a dimension reduction method to partition firms into classes. We now describe a computationally tractable two-step grouped fixed-effects estimator, where we classify firms in a first step and estimate earnings and mobility parameters in a second step.

4.1. *Recovering Firm Classes Using k-Means Clustering*

In both the static and dynamic models described in Section 2, the distributions of log-earnings Y_{it} and characteristics X_{it}, and the probabilities of mobility m_{it}, are all allowed to depend on firm classes k, but not on the identity of the firm within class k. In other words, unobservable firm heterogeneity operates at the level of firm classes in the model, not at the level of individual firms. For example, in (8), the first period's distribution of log-earnings in firm j does not depend on j beyond its dependence on firm class $k = k(j)$:

$$\Pr[Y_{i1} \leq y_1 | j_{i1} = j] = \sum_{\alpha=1}^{L} F_{k\alpha}(y_1) q_k(\alpha), \tag{11}$$

where the left-hand side thus only depends on $k = k(j)$. This observation motivates classifying firms into classes using their earnings distributions, as we now explain.

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We propose partitioning the J firms in the sample into classes by solving the following weighted k-means problem:

$$\min_{k(1),\ldots,k(J),H_1,\ldots,H_K} \sum_{j=1}^{J} n_j \int \left(\widehat{F}_j(y) - H_{k(j)}(y)\right)^2 d\mu(y), \tag{12}$$

where $\widehat{F}_j$ denotes the empirical cdf of log-earnings in firm j, n_j is the number of workers in firm j, μ is a discrete or continuous measure (μ is supported on a finite grid in our application), $k(1), \ldots, k(J)$ denotes a partition of firms into K classes, and $H_1, \ldots, H_K$ are cdfs. Here, we take the number K of firm classes as known. In the empirical analysis, we will check robustness around the baseline value $K = 10$. We minimize (12) with respect to all possible partitions and to class-specific cdfs. While global minima in k-means may be challenging to compute, k-means algorithms are widely used in many fields and efficient heuristic computational methods have been developed (e.g., Steinley (2006)).

Through the classification in (12), we estimate firm classes as "discrete fixed effects," allowing them to be correlated to firm-specific covariates. In our application on short panels, we will assume that the firms' classification is time-invariant, and we will correlate the estimated classes ex post to firm observables.

To provide a formal justification for the classification, we consider a setting where the model (either static or dynamic) is well-specified, and there exists a partition of firms into K classes in the population. We consider an asymptotic sequence where both the number of firms and the number of workers per firm tend to infinity. Using a result from Bonhomme and Manresa (2015), we show that estimated firm classes, $\widehat{k}(j)$, converge to the population ones up to an arbitrary labeling as the sample size grows. As a result, the asymptotic distribution of parameter estimates in the second step is not affected by the estimation of firm classes. In Appendix B, we provide details on this analysis.

Two remarks are in order. First, the labeling of the classes is arbitrary. This labeling does not affect variance decomposition exercises. However, a structural interpretation of the firm classes in terms of productivity would require additional assumptions. Eeckhout and Kircher (2011) showed that in some cases, it can be impossible to recover such a productivity ordering from wage data only.

Second, the classification fails to be identified when two firm classes have identical earnings distributions in the cross-section. This can happen if one firm offers a higher earnings schedule but has lower-type workers, and another one offers a lower earnings schedule but has higher-type workers, in such a way that the two firms have exactly the same earnings distributions. In some environments without firm capacity constraints, such as Postel-Vinay and Robin (2002), the upper bound of earnings in the firm is increasing in firm productivity, so firm-specific distributions are all different and firms may be consistently classified based on their earnings distributions. It is difficult to obtain similar guarantees in models with capacity constraints. Nevertheless, note that identification can survive even when some firms have identical mean earnings, provided the *distributions* of earnings differ.

In the empirical analysis, we attempt to bring additional information beyond cross-sectional earnings to learn about firm heterogeneity, in several ways. Going beyond wage information has been advocated in the structural literature (e.g., Eeckhout and Kircher (2011), Bagger and Lentz (2014), Hagedorn, Law, and Manovskii (2017), Bartolucci, Devicienti, and Monzon (2015)). As robustness checks, we use reclassification and random-effects methods, which rely on longitudinal information on both earnings and mobility.

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We also experiment, albeit without a structural model, with bringing other information on the firm in order to inform the classification, such as firm value added and measures of worker flows.

4.2. *Recovering the Model's Parameters in a Second Step*

From the first step, we obtain estimates of firm classes $\widehat{k}(j)$ for all firms j in the sample. Given those, we then impute a class $\widehat{k}_{it} = \widehat{k}(j_{it})$ to each worker-period observation, and we estimate the model conditional on the $\widehat{k}_{it}$'s. To describe this second step, we consider a specification where workers belong to L latent types, and the model is parametric given worker and firm heterogeneity. We focus on a two-period version of the static model, and a four-period version of the dynamic model, both of which we estimate on Swedish data.

In the static case, we let $f_{k\alpha}(y; \theta_f)$ (first-period earnings), $f^m_{k\alpha}(y; \theta_{fm})$ (second-period earnings for job movers), $q_k(\alpha; \theta_q)$ (worker-type proportions), and $p_{kk'}(\alpha; \theta_p)$ (worker-type proportions for job movers) be indexed by parameter vectors θ_f, θ_{fm}, θ_q, θ_p. In our baseline specification, we will let both earnings densities be log-normal with (k, α)-specific means and variances. That is, means and variances of log-earnings are allowed to differ between all combinations of worker types and firm classes. In addition, in the time dimension, we will allow for full interactions between firm classes and time indicators, as well as unrestricted non-stationary variances. Last, we will treat all $q_k(\alpha)$ and $p_{kk'}(\alpha)$ as unrestricted parameters.

Following the spirit of the identification strategy, we first estimate log-earnings densities using job movers only, and we then estimate worker-type proportions in the first period using both job movers and job stayers.[9] Under the assumption that worker types and earnings realizations are independent across workers conditional on mobility indicators and firm classes, the log-likelihood of job movers conditional on mobility patterns and estimated firm classes takes the following form (N_m denoting the number of job movers):

$$\sum_{i=1}^{N_m}\sum_{k=1}^{K}\sum_{k'=1}^{K}\mathbf{1}\{\widehat{k}_{i1}=k\}\mathbf{1}\{\widehat{k}_{i2}=k'\}\ln\left(\sum_{\alpha=1}^{L}p_{kk'}(\alpha;\theta_p)f_{k\alpha}(Y_{i1};\theta_f)f^m_{k'\alpha}(Y_{i2};\theta_{fm})\right). \tag{13}$$

In turn, the log-likelihood of all workers in period 1 is

$$\sum_{i=1}^{N}\sum_{k=1}^{K}\mathbf{1}\{\widehat{k}_{i1}=k\}\ln\left(\sum_{\alpha=1}^{L}q_k(\alpha;\theta_q)f_{k\alpha}(Y_{i1};\widehat{\theta}_f)\right). \tag{14}$$

Hence, conditional on the estimated firm classes, (13) and (14) are standard single-agent correlated random-effects log-likelihood functions. We estimate $\widehat{\theta}_f$, $\widehat{\theta}_{fm}$, $\widehat{\theta}_p$ by maximizing (13), and then $\widehat{\theta}_q$ by maximizing (14). We use the EM algorithm for computation (Dempster, Laird, and Rubin (1977)).

Dynamic Model

We use a similar approach for the dynamic finite mixture model using four periods; see equations (9) and (10). In this case, we specify the conditional mean of Y_{i4} given Y_{i3} and

[9]Proceeding in this way has the advantage of recovering earnings parameters from job movements directly, albeit with some efficiency loss. In practice, we estimate the type proportions of job stayers in the last step, and combine them with the $p_{kk'}(\alpha)$ to recover the unconditional proportions $q_k(\alpha)$.

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worker and firm heterogeneity as $\mu_{4k'\alpha} + \rho_{4|3} Y_{i3}$, where $\mu_{4k'\alpha}$ is a (k', α)-specific intercept. Likewise, the conditional mean of Y_{i1} given Y_{i2} and worker and firm heterogeneity is $\mu_{1k\alpha} + \rho_{1|2} Y_{i2}$. The parameters $\rho_{4|3}$ and $\rho_{1|2}$ capture the persistence of log-earnings within job. For parsimony, we have imposed that those parameters are homogeneous across worker types and firm classes, although this could be relaxed with a larger sample.

In addition, we specify the mean of (Y_{i2}, Y_{i3}) for job movers between classes k and k' as $(\mu_{2k\alpha} + \xi_2(k'), \mu_{3k'\alpha} + \xi_3(k))$. The term $\xi_2(k')$ reflects that, conditional on moving between k and k', mean log-earnings before the move can differ with the firm of destination, due to the presence of *endogenous mobility*. The term $\xi_3(k)$ reflects that the previous firm is allowed to have a direct effect on log-earnings after a move, through the presence of *state dependence*. Neither of those effects is allowed for in the static version of the model. We specify the mean of (Y_{i2}, Y_{i3}) for job stayers in a firm of class k as $(\mu^s_{2k\alpha}, \mu^s_{3k\alpha})$. Last, for both stayers and movers, we let the covariance matrices vary with firm classes.[10]

Given estimates $\widehat{\rho}_{4|3}$ and $\widehat{\rho}_{1|2}$ of the persistence parameters, the other parameters can be estimated using a similar approach as in the static case, based on the following log-likelihood functions:

$$\sum_{i=1}^{N_m} \sum_{k=1}^{K} \sum_{k'=1}^{K} \mathbf{1}\{\widehat{k}_{i2} = k\} \mathbf{1}\{\widehat{k}_{i3} = k'\} \times \cdots$$
$$\times \ln\left(\sum_{\alpha=1}^{L} p_{kk'}(\alpha; \theta_p) f^f_{Y_{i2},k\alpha}(Y_{i1}; \widehat{\rho}_{1|2}, \theta_{f^f}) f^m_{kk'\alpha}(Y_{i2}, Y_{i3}; \theta_{f^m}) f^b_{Y_{i3},k'\alpha}(Y_{i4}; \widehat{\rho}_{4|3}, \theta_{f^b}) \right), \quad (15)$$

and

$$\sum_{i=1}^{N} \sum_{k=1}^{K} \mathbf{1}\{\widehat{k}_{i2} = k\}$$
$$\times \ln\left(\sum_{\alpha=1}^{L} q_k(\alpha; \theta_q) f^f_{Y_{i2}k\alpha}(Y_{i1}; \widehat{\rho}_{1|2}, \widehat{\theta}_{f^f}) f^s_{k\alpha}(Y_{i2}, Y_{i3}; \theta_{f^s}) f^b_{Y_{i3},k'\alpha}(Y_{i4}; \widehat{\rho}_{4|3}, \widehat{\theta}_{f^b}) \right). \quad (16)$$

We estimate $\widehat{\theta}_p$, $\widehat{\theta}_{f^f}$, $\widehat{\theta}_{f^m}$, $\widehat{\theta}_{f^b}$ based on (15), and then $\widehat{\theta}_q$, $\widehat{\theta}_{f^s}$ based on (16), using the EM algorithm in both cases.

While it is in principle possible to estimate $\rho_{4|3}$ and $\rho_{1|2}$ by maximizing a joint likelihood function across movers and stayers with respect to all parameters, doing so would be computationally cumbersome. A convenient alternative, which we adopt in the empirical analysis, is to estimate these parameters in an initial step based on covariance restrictions. Under the assumption that the effect of worker types on mean log-earnings is constant over time within firm, simple restrictions on the ρ's can be obtained by exploiting the particular form of the conditional means of Y_{i4} given Y_{i3} and Y_{i1} given Y_{i2}, respectively. We provide details on the covariance-based estimation of $\rho_{4|3}$ and $\rho_{1|2}$ in the Supplemental Material.

The estimation approach outlined in this section can be modified in several ways that we implement empirically. A first extension is a model-based reclassification. Given estimates of the θ parameters, one may reclassify every firm j to the class $k = \widetilde{k}(j)$ that

[10]We impose that the best linear predictors in the regressions of Y_{i3} on Y_{i2}, for both stayers (denoted as $\rho^s_{3|2}$) and movers ($\rho^m_{3|2}$), do not depend on worker types or firm classes, and that the residual variances in the case of movers only depend on k'. This could be relaxed with a large enough sample.

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